

Algebraic Geometry – Exercise Sheet 1

Week 1: Affine algebraic sets, projective space and localisation
Monday and Thursday afternoon sessions; Lectures 1–5

Difficulty. ★ routine; ★★ synthesis; ★★★ challenge or extension.

Use. The sheet is deliberately longer than the exercise sessions. Cerulean exercises form the suggested route; green exercises are alternatives.

Monday afternoon — material available: Lecture 1 only

Exercise 1. Three points in the affine plane

★ CORE

Let

$$S = \{(0, 0), (1, 0), (0, 1)\} \subset \mathbb{k}^2.$$

(i) Show that

$$\mathbf{I}(S) = (x, y) \cap (x - 1, y) \cap (x, y - 1).$$

(ii) Prove directly that

$$\mathbf{I}(S) = (xy, x(x - 1), y(y - 1)).$$

(iii) Show that evaluation at the three points induces an isomorphism

$$\mathbb{k}[x, y]/\mathbf{I}(S) \simeq \mathbb{k}^3.$$

Exercise 2. Two axes and their coordinate rings

★ CORE

In $\mathbb{k}^2$, set $X = \mathbf{V}(x)$ and $Y = \mathbf{V}(y)$.

(i) Show that

$$X \cup Y = \mathbf{V}(xy), \quad X \cap Y = \mathbf{V}(x, y).$$

(ii) Compute the coordinate rings of X , Y , $X \cup Y$, and $X \cap Y$.

(iii) Explain geometrically why $\mathbb{k}[x, y]/(xy)$ has zero divisors.

Exercise 3. The parabola and the cusp

★★ CORE

Consider

$$P = \mathbf{V}(y - x^2), \quad C = \mathbf{V}(y^2 - x^3) \subset \mathbb{k}^2.$$

(i) Show that $t \mapsto (t, t^2)$ defines an isomorphism $\mathbb{A}^1 \rightarrow P$.

(ii) Show that $t \mapsto (t^2, t^3)$ defines a morphism $\nu : \mathbb{A}^1 \rightarrow C$ which is bijective on the underlying sets of $\mathbb{k}$ -points.

(iii) Write the corresponding homomorphism of coordinate rings and prove that ν is not an isomorphism.

Exercise 4. Polynomial maps of the affine line

★ CHOICE

Show that morphisms of affine algebraic sets $\mathbb{A}_{\mathbb{k}}^1 \rightarrow \mathbb{A}_{\mathbb{k}}^1$ are exactly polynomial maps $t \mapsto f(t)$. Describe the coordinate-ring homomorphism and check explicitly how composition reverses arrows.

Thursday afternoon — material available: Lectures 1–4

Exercise 5. Radicals and reduced rings

★ CORE

Compute

$$\sqrt{(x^2, y^3)}, \quad \sqrt{(x^2 y^3)}$$

in $\mathbb{k}[x, y]$. Decide which of the rings

$$\mathbb{k}[x]/(x^2), \quad \mathbb{k}[x, y]/(xy), \quad \mathbb{k}[x, y]/(x^2, xy)$$

are reduced.

Exercise 6. Points at infinity

★ CORE

In the affine chart $x_0 \neq 0$ of $\mathbb{P}_{\mathbb{k}}^2$, write $x = x_1/x_0$ and $y = x_2/x_0$.

- (i) Homogenise the parabola $y = x^2$ and determine its points at infinity.
- (ii) Homogenise the hyperbola $xy = 1$ and determine its points at infinity.
- (iii) Explain geometrically why the answers are different.

Exercise 7. The line through two projective points

★ **CHOICE**

Let $p = [a_0 : \cdots : a_n]$ and $q = [b_0 : \cdots : b_n]$ be distinct points of $\mathbb{P}_{\mathbb{K}}^n$. Show that the line through them is defined by the 3×3 minors of

$$\begin{pmatrix} a_0 & \cdots & a_n \\ b_0 & \cdots & b_n \\ x_0 & \cdots & x_n \end{pmatrix}.$$

Write the equation explicitly for $n = 2$.

Exercise 8. Points in $\mathbb{P}^2$ from minors

★★ **CHOICE**

Show that the 2×2 minors of

$$M = \begin{pmatrix} x & 0 \\ y & y \\ 0 & z \end{pmatrix}$$

define the three coordinate points in $\mathbb{P}^2$. For four lines $\ell_1, \dots, \ell_4$ with no three concurrent, show that the maximal minors of

$$\begin{pmatrix} \ell_1 & 0 & 0 \\ -\ell_2 & \ell_2 & 0 \\ 0 & -\ell_3 & \ell_3 \\ 0 & 0 & -\ell_4 \end{pmatrix}$$

define their six pairwise intersection points.

Exercise 9. Equations of the Segre and Veronese images

★★★ **OPTIONAL**

a) For

$$([s_0 : s_1], [t_0 : t_1]) \mapsto [s_0 t_0 : s_0 t_1 : s_1 t_0 : s_1 t_1],$$

compute the kernel of $z_{ij} \mapsto s_i t_j$ and identify the image in $\mathbb{P}^3$.

b) For the quadratic Veronese map $\mathbb{P}^2 \rightarrow \mathbb{P}^5$, arrange the target coordinates as a symmetric 3×3 matrix and show that the image is defined by its 2×2 minors.

Exercise 10. Localising the coordinate axes

★ **CORE**

Let

$$A = \mathbb{k}[x, y]/(xy).$$

- (i) Compute A_x and A_y .
- (ii) Compute $A_{(x)}$ and $A_{(y)}$.
- (iii) Explain what these localisations retain and what they discard geometrically.

Exercise 11. Localisation with a nilpotent

★ **CHOICE**

Let $A = \mathbb{k}[\varepsilon]/(\varepsilon^2)$. Compute

$$A_\varepsilon, \quad A_{(\varepsilon)}, \quad A_{1+\varepsilon}.$$

Explain why the three answers are so different.

Exercise 12. Two finite module structures

★★ **CHOICE**

Let

$$B = \mathbb{k}[x, y]/(y^2 - x).$$

Show that B is free of rank 2 over $\mathbb{k}[x]$ and free of rank 1 over $\mathbb{k}[y]$. Compare geometrically the corresponding morphisms to the affine line.

Exercise 13. Noether normalisation for the hyperbola

★★★ **CHOICE**

Let

$$B = \mathbb{k}[x, y]/(xy - 1), \quad u = x + y \in B.$$

Show that B is finite over $\mathbb{k}[u]$. Deduce that the corresponding morphism to $\mathbb{A}_{\mathbb{k}}^1$ has fibres with at most two points.

After Friday's lecture or homework

Exercise 14. A presheaf which is not a sheaf

★ OPTIONAL

Let $X = U \amalg V$ be the disjoint union of two non-empty open subsets. Define a presheaf $\mathcal{F}$ by

$$\mathcal{F}(W) = \mathbb{Z} \quad \text{for } W \neq \emptyset, \quad \mathcal{F}(\emptyset) = 0,$$

with the evident restriction maps. Show that $\mathcal{F}$ is not a sheaf.

Exercise 15. Continuous and holomorphic functions

★★ CHOICE

Let $U \subset \mathbb{C}$ be connected and contain at least two points. Show that $\mathcal{O}(U)$ has no zero divisors, whereas $\mathcal{C}^0(U)$ has non-zero functions f, g with $fg = 0$.

Exercise 16. $\mathcal{O}(d)$ on the projective line

★★★ OPTIONAL

Identify $\mathbb{P}_{\mathbb{C}}^1$ with $\mathbb{C} \cup \{\infty\}$. Define $\mathcal{O}(d)$ as the sheaf of meromorphic functions which are holomorphic away from ∞ and have at most a pole of order d there when $d \geq 0$; for $d < 0$, require a zero of order at least $-d$ at ∞ . Compute its global sections.

Algebraic Geometry – Exercise Sheet 2

Week 2: Prime spectra, the structure sheaf and affine subschemes
Monday and Thursday afternoon sessions; Lectures 6–10

Difficulty. ★ routine; ★★ synthesis; ★★★ challenge or extension.

Use. The sheet is deliberately longer than the exercise sessions. Cerulean exercises form the suggested route; green exercises are alternatives.

Monday afternoon — material available: Lecture 6 only

Exercise 1. The affine line as a spectrum

★ CORE

Assume that $\mathbb{k}$ is algebraically closed and let $X = \operatorname{Spec}(\mathbb{k}[t])$.

- (i) List the points of X .
- (ii) Determine which points are closed and compute the closure of each point.
- (iii) For a non-zero polynomial f , describe the closed subset $\mathbb{V}(f)$.

Exercise 2. The spectrum of the integers

★ CORE

Describe $\operatorname{Spec}(\mathbb{Z})$: its points, its closed points, and the closure of each point. For $n \neq 0$, determine $\mathbb{V}(n)$.

Exercise 3. Nilpotents are invisible topologically

★ CORE

Let

$$A = \mathbb{k}[\varepsilon]/(\varepsilon^2), \quad X = \operatorname{Spec}(A).$$

Determine the points and open subsets of X . Compare the underlying topological space with $\operatorname{Spec}(\mathbb{k})$, and explain why the two affine schemes are nevertheless different.

Exercise 4. A quotient map on spectra

★ CHOICE

Consider the quotient homomorphism

$$\mathbb{k}[t] \longrightarrow \mathbb{k}[t]/(t^2).$$

Describe the induced map on spectra and its image. Why does the image depend only on $\sqrt{(t^2)}$?

Exercise 5. Closed subsets in the node

★★ CHOICE

Let $A = \mathbb{k}[x, y]/(xy)$ and $X = \operatorname{Spec}(A)$. Denote the classes of x, y by the same letters.

- (i) Show that $\mathbb{V}(x) \cup \mathbb{V}(y) = X$ and determine $\mathbb{V}(x) \cap \mathbb{V}(y)$.
- (ii) Compute the closures of the points (x) and (y) .
- (iii) Show that neither of these two points is closed.

Exercise 6. The affine line over a function field

★★ CHOICE

Let $K = \mathbb{C}(t)$ and let $X = \operatorname{Spec}(K[x])$.

- (i) Describe the points and the closed points of X .
- (ii) Which closed points are K -rational? Give a closed point which is not K -rational.
- (iii) Put $S = \mathbb{C}[t] \setminus \{0\}$. Use $K[x] = S^{-1}\mathbb{C}[t, x]$ to identify X with the prime ideals of $\mathbb{C}[t, x]$ lying over $(0) \in \operatorname{Spec}(\mathbb{C}[t])$.

Thursday afternoon — material available: Lectures 6–9

Exercise 7. Principal opens in the node

★★ CORE

Let $A = \mathbb{k}[x, y]/(xy)$ and $X = \operatorname{Spec}(A)$.

- (i) Describe $D(x)$ and $D(y)$ as affine schemes.
- (ii) Show that $D(x) \cap D(y) = \emptyset$ and identify the point missing from $D(x) \cup D(y)$.
- (iii) Compute the residue fields at (x) , (y) and (x, y) .

Exercise 8. Sections and a stalk on the affine line★ **CORE**

Let $X = \operatorname{Spec}(\mathbb{k}[t])$ and $U = D(t(t-1))$.

- (i) Compute $\mathcal{O}_X(U)$.
- (ii) Compute the stalk $\mathcal{O}_{X,(t-a)}$ and its residue field.
- (iii) For which a does $(t-a)$ belong to U ?

Exercise 9. A subscheme of the affine line with two support points★★ **CORE**

Let

$$Z = \operatorname{Spec}(\mathbb{k}[t]/(t^3(t-1)^2)).$$

- (i) Determine the underlying topological space of Z .
- (ii) Decompose its coordinate ring as a product of two rings.
- (iii) Compare the two local pieces with the reduced points at 0 and 1.

Exercise 10. Scheme-theoretic unions and intersections★ **CORE**

In $\mathbb{A}_{\mathbb{k}}^2$, let

$$X = \mathbb{V}(x), \quad Y = \mathbb{V}(y), \quad Y' = \mathbb{V}(y^2).$$

Compute the ideals defining $X \cup Y$, $X \cap Y$, $X \cup Y'$ and $X \cap Y'$. Compare their underlying closed subsets.

Exercise 11. Products of rings and disjoint unions★★ **CHOICE**

Let B and C be non-zero rings.

- (i) Show that every prime ideal of $B \times C$ is of the form $\mathfrak{p} \times C$ or $B \times \mathfrak{q}$.
- (ii) Deduce that

$$\operatorname{Spec}(B \times C) \simeq \operatorname{Spec}(B) \amalg \operatorname{Spec}(C).$$

- (iii) Relate this to the idempotents $(1, 0)$ and $(0, 1)$.

After Friday's arithmetic lecture or homework**Exercise 12. Prime ideals in $\mathbb{Z}[\sqrt{3}]$** ★★ **CHOICE**

Let

$$A = \mathbb{Z}[s]/(s^2 - 3), \quad f : \operatorname{Spec}(A) \longrightarrow \operatorname{Spec}(\mathbb{Z}).$$

For a prime p , describe the points of $\operatorname{Spec}(A)$ mapping to (p) by factoring $s^2 - 3$ over $\mathbb{F}_p$. Treat separately $p = 2$, $p = 3$, and $p \neq 2, 3$.

Exercise 13. An arithmetic conic★★★ **OPTIONAL**

For

$$X = \operatorname{Spec}(\mathbb{Z}[x, y]/(x^2 + y^2 - 5)) \longrightarrow \operatorname{Spec}(\mathbb{Z}),$$

write down the equation obtained after reduction modulo a prime p . Determine for which p that equation is a square or has a singular point.

Algebraic Geometry – Exercise Sheet 3

Week 3: Schemes, morphisms, gluing and first global properties
Monday and Thursday afternoon sessions; Lectures 11–15

Difficulty. ★ routine; ★★ synthesis; ★★★ challenge or extension.

Use. The sheet is deliberately longer than the exercise sessions. Cerulean exercises form the suggested route; green exercises are alternatives.

Monday afternoon — material available: Lectures 1–11

Exercise 1. Global functions as morphisms to the affine line

★ **CORE**

Let X be a scheme.

- (i) Explain why a global section $s \in \Gamma(X, \mathcal{O}_X)$ determines a morphism

$$f_s : X \longrightarrow \mathbb{A}_{\mathbb{Z}}^1.$$

- (ii) Show that

$$\mathrm{Hom}(X, \mathbb{A}_{\mathbb{Z}}^1) \simeq \Gamma(X, \mathcal{O}_X).$$

- (iii) If $X = \mathrm{Spec}(A)$, describe the morphism associated with $a \in A$.

Exercise 2. The reduced point and the double point

★ **CORE**

Inside $\mathbb{A}_{\mathbb{k}}^1 = \mathrm{Spec}(\mathbb{k}[t])$, consider

$$P = \mathrm{Spec}(\mathbb{k}[t]/(t)), \quad D = \mathrm{Spec}(\mathbb{k}[t]/(t^2)).$$

- (i) Show that the two closed immersions have the same image as subsets of $\mathbb{A}_{\mathbb{k}}^1$.
(ii) Show that P and D are not isomorphic as schemes.
(iii) Compute the maps on local rings at their unique points.

Exercise 3. A morphism between affine schemes

★ **CORE**

Let

$$\varphi : \mathbb{k}[u, v] \longrightarrow \mathbb{k}[t], \quad u \longmapsto t, \quad v \longmapsto t^2.$$

- (i) Describe the induced morphism $f : \mathbb{A}_{\mathbb{k}}^1 \rightarrow \mathbb{A}_{\mathbb{k}}^2$.
(ii) Determine $\ker(\varphi)$ and factor f through a closed subscheme of $\mathbb{A}_{\mathbb{k}}^2$.
(iii) Show that the resulting morphism onto that closed subscheme is an isomorphism.

Exercise 4. An arithmetic spectrum

★★ **CHOICE**

Let $A = \mathbb{Z}[s]/(s^2 - 3)$. Describe the inverse image in $\mathrm{Spec}(A)$ of the points (2) , (3) , and (p) for $p \neq 2, 3$, according to the factorisation of $s^2 - 3$ modulo p .

Thursday afternoon — material available: Lectures 1–14

Exercise 5. Dual numbers and tangent vectors

★★ **CORE**

Let $D = \mathrm{Spec}(\mathbb{k}[\varepsilon]/(\varepsilon^2))$ and let $p = (a, b) \in \mathbb{A}_{\mathbb{k}}^2(\mathbb{k})$.

- (i) Show that morphisms $D \rightarrow \mathbb{A}_{\mathbb{k}}^2$ whose closed point maps to p are uniquely of the form

$$x \longmapsto a + u\varepsilon, \quad y \longmapsto b + v\varepsilon.$$

- (ii) Let $C = \mathbb{W}(f) \subset \mathbb{A}_{\mathbb{k}}^2$. Show that such a morphism factors through C precisely when

$$f(a, b) = 0, \quad f_x(a, b)u + f_y(a, b)v = 0.$$

- (iii) Apply this to the parabola $y = x^2$.

Exercise 6. The line with two origins★★ **CORE**

Glue two copies

$$X_1 = \operatorname{Spec}(\mathbb{k}[t_1]), \quad X_2 = \operatorname{Spec}(\mathbb{k}[t_2])$$

along $D(t_1) \simeq D(t_2)$ by $t_1 \mapsto t_2$. Let X be the resulting scheme.

- (i) Describe the underlying topological space.
- (ii) Show that $\Gamma(X, \mathcal{O}_X) \simeq \mathbb{k}[t]$.
- (iii) Show that the canonical map $X \rightarrow \operatorname{Spec} \Gamma(X, \mathcal{O}_X)$ identifies the two origins.
- (iv) Deduce that X is not affine.

Exercise 7. Global functions on the projective line★★ **CHOICE**Construct $\mathbb{P}_{\mathbb{k}}^1$ by gluing $U_0 = \operatorname{Spec}(\mathbb{k}[t])$ and $U_1 = \operatorname{Spec}(\mathbb{k}[s])$ along $s = t^{-1}$. Prove that

$$\Gamma(\mathbb{P}_{\mathbb{k}}^1, \mathcal{O}_{\mathbb{P}^1}) = \mathbb{k}.$$

Exercise 8. The projective conic from its standard opens★★ **CORE**

Let

$$S = \mathbb{k}[x_0, x_1, x_2]/(x_0x_2 - x_1^2), \quad C = \operatorname{Proj}(S).$$

- (i) Compute $(S_{x_0})_0$ and $(S_{x_2})_0$.
- (ii) Identify $D_+(x_0)$ and $D_+(x_2)$ with affine lines.
- (iii) Describe their overlap and recover the gluing rule for $\mathbb{P}^1$.

Exercise 9. Connectedness and idempotents★★ **CORE**

- (i) Prove that $\operatorname{Spec}(A)$ is disconnected if and only if A has a non-trivial idempotent.
- (ii) Apply this to $A = \mathbb{k}[t]/(t^2 - t)$.
- (iii) Show that $\operatorname{Spec}(\mathbb{k}[x, y]/(xy))$ is connected although reducible.

Exercise 10. Reduction and its universal property★★ **CORE**

Let

$$A = \mathbb{k}[x, y]/(x^2, xy), \quad X = \operatorname{Spec}(A).$$

- (i) Compute the nilradical and X_{red} .
- (ii) Let R be a reduced $\mathbb{k}$ -algebra. Show that every homomorphism $A \rightarrow R$ factors uniquely through A_{red} .
- (iii) Translate this into a statement about morphisms of schemes.

Exercise 11. A weighted projective chart★★★ **OPTIONAL**Let $R = \mathbb{k}[x, y, z]$ with $\deg x = \deg y = 1$ and $\deg z = 2$, and set $X = \operatorname{Proj}(R)$. Show that $D_+(x)$ and $D_+(y)$ are affine planes and that

$$D_+(z) \simeq \operatorname{Spec}(\mathbb{k}[a, b, c]/(ac - b^2)).$$

Identify the singular point of this chart.

After Friday's lecture or preparation for Week 4**Exercise 12. Components and generic points**★★ **CHOICE**

Let

$$I = (xy, xz) \subset \mathbb{k}[x, y, z], \quad X = \operatorname{Spec}(\mathbb{k}[x, y, z]/I).$$

Show that $I = (x) \cap (y, z)$. Determine the irreducible components, their generic points and dimensions, and decide whether X is connected or integral.**Exercise 13. The function field of the cusp**★★ **CHOICE**

Let

$$C = \operatorname{Spec}(\mathbb{k}[x, y]/(y^2 - x^3)).$$

Show that C is integral and that $K(C) \simeq \mathbb{k}(t)$.

Algebraic Geometry – Exercise Sheet 4

Week 4: Generic points, dimension and scheme-theoretic fibres
One Monday exercise session, followed by a revision bank; Lectures 15–18

Difficulty. ★ routine; ★★ synthesis; ★★★ challenge or extension.

Use. The sheet is deliberately longer than the exercise sessions. Cerulean exercises form the suggested route; green exercises are alternatives.

Monday afternoon — material available: Lectures 1–16

Exercise 1. Components and generic points

★★ CORE

Let

$$I = (xy, xz) \subset \mathbb{k}[x, y, z], \quad X = \operatorname{Spec}(\mathbb{k}[x, y, z]/I).$$

- (i) Show that $I = (x) \cap (y, z)$.
- (ii) Determine the irreducible components and their generic points.
- (iii) Is X reduced, connected, irreducible, or integral?

Exercise 2. The function field of the cusp

★★ CORE

Let

$$C = \operatorname{Spec}(\mathbb{k}[x, y]/(y^2 - x^3)).$$

Show that C is integral, compute its function field, and deduce its dimension.

Exercise 3. Chains and dimensions

★ CORE

- (i) Exhibit a maximal chain of irreducible closed subsets in $\operatorname{Spec}(\mathbb{Z})$.
- (ii) Give a chain of length n in $\mathbb{A}_{\mathbb{k}}^n$.
- (iii) Determine the dimension of

$$\operatorname{Spec}(\mathbb{k}[x, y, z]/(xy, xz))$$

from its components.

Exercise 4. Noether normalisation and dimension

★★ CORE

Let

$$B = \mathbb{k}[x, y]/(y^2 - x^3).$$

- (i) Show that B is finite over $\mathbb{k}[x]$.
- (ii) Use this to recover $\dim(B) = 1$.
- (iii) Compare with the computation from the function field.

After Lectures 17–18 — revision bank on fibres

Exercise 5. The generic affine line

★ CORE

Let

$$\pi : \mathbb{A}_{\mathbb{C}}^2 = \operatorname{Spec}(\mathbb{C}[t, x]) \longrightarrow \mathbb{A}_{\mathbb{C}}^1 = \operatorname{Spec}(\mathbb{C}[t])$$

be the projection. Compute its fibre over a closed point $(t - a)$ and its fibre over the generic point (0) . Explain why the latter is $\operatorname{Spec}(\mathbb{C}(t)[x])$ and verify the dimension formula for the generic fibre.

Exercise 6. A fibre product larger than the set-theoretic one

★★ CHOICE

Let $X = \operatorname{Spec}(\mathbb{C})$ and $S = \operatorname{Spec}(\mathbb{R})$. Compute $X \times_S X$ and compare its points with the set-theoretic fibre product of the underlying one-point spaces.

Exercise 7. The power map and its fibres

★★ CORE

Assume that $\mathbb{k}$ is algebraically closed of characteristic 0. Let

$$f : \mathbb{A}_{\mathbb{k}}^1 \longrightarrow \mathbb{A}_{\mathbb{k}}^1, \quad t \longmapsto t^n.$$

Compute every scheme-theoretic fibre, determine when it is reduced, compute the generic fibre, and show that f is finite.

Exercise 8. A family whose special fibre becomes reducible

★★ **CORE**

Let

$$X = \operatorname{Spec}(\mathbb{k}[x, y, t]/(xy - t)) \longrightarrow \mathbb{A}_t^1.$$

Compute the fibres, show that all have dimension 1, and determine which are integral and which are reducible.

Exercise 9. A jumping fibre

★★ **CORE**

Consider

$$f : \mathbb{A}_{\mathbb{k}}^2 \longrightarrow \mathbb{A}_{\mathbb{k}}^2, \quad (x, y) \longmapsto (x, xy).$$

Compute the fibre over (a, b) and show that the generic fibre has dimension 0, while the fibre over $(0, 0)$ has dimension 1.

Exercise 10. The generic fibre formula in examples

★★ **CHOICE**

Verify

$$\dim X_\eta = \dim X - \dim Y$$

for the map in the preceding exercise and for the projection $\mathbb{A}_{\mathbb{k}}^{m+n} \rightarrow \mathbb{A}_{\mathbb{k}}^m$.

Optional outreach exercises: Hilbert schemes of points

Exercise 11. $\operatorname{Hilb}^n(\mathbb{A}^1)$

★★ **CHOICE**

Assume that $\mathbb{k}$ is algebraically closed. Show that every ideal $I \subset \mathbb{k}[x]$ with $\dim_{\mathbb{k}} \mathbb{k}[x]/I = n$ is generated by a unique monic polynomial of degree n . Deduce

$$\operatorname{Hilb}^n(\mathbb{A}_{\mathbb{k}}^1) \simeq \mathbb{A}_{\mathbb{k}}^n$$

and explain how factorisation records support and multiplicities.

Exercise 12. Commuting matrices and a cyclic vector

★★★ **OPTIONAL**

Let V be n -dimensional and let (X, Y, v) satisfy $[X, Y] = 0$ and $\mathbb{k}[X, Y]v = V$. Show that

$$I = \{f \in \mathbb{k}[x, y] \mid f(X, Y)v = 0\}$$

is an ideal and $\mathbb{k}[x, y]/I \simeq V$. Determine the triple for $I = (y, x^2)$.